

Result: 100% 20 / 20	 101810 000751	Neptun ID: Draft papers: ☐ Name: _____
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The test consists of **5 questions**. The available time to solve the quiz is **20 minutes**. Please use a blue coloured pen; the use of any other tools is forbidden.

- | | a | b | c | d | e | f |
|-----|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| 1.) | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |
| 2.) | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |
| 3.) | ☐ | ☐ | ☐ | ☐ | | |
| 4.) | ☐ | ☐ | ☐ | ☐ | | |
| 5.) | (essay) | | | | | |

Please copy the multiple choice questions' solution in the attached table. For each correct answer marked, 2 points are awarded; for each incorrect answer marked, -1 point is awarded. There is no limit to the number of correct answers per question. A negative aggregate score will be interpreted as 0 points.

⊕

1.) Let $L^{(0)} = \langle LC, Con, Form \rangle$ be a zero-order language such that $Con = \{p, q\}$. Which string of symbols are membered in the set $Form$?

2 $\neg(\neg p \supset \neg q)$

(b) $((p))$

(c) $\neg\neg r$

—

2 $(p \equiv \neg q)$

2 $\neg\neg q$

(f) $(p \subset q)$

⊕

2.) Which statements are true for the formula below?

$$((\neg q \equiv p) \supset \neg(p \vee (\neg q \wedge p)))$$

(a) $(p \equiv \neg q)$ is a direct subformula

(b) $(\neg q \wedge p)$ is a direct subformula

(c) $((\neg q \equiv p) \supset \neg(p \vee (\neg q \wedge p)))$ is a direct subformula

2 $((\neg q \equiv p) \supset \neg(p \vee (\neg q \wedge p)))$ is a subformula

2 $(\neg q \equiv p)$ is a subformula

2 $(\neg q \wedge p)$ is a subformula

3.) Which interpretations are models of the following set of formulae?

$$\{ (\neg p \vee q), (q \vee (p \vee (q \vee q))) \}$$

(a) $\varrho(p) = 0, \quad \varrho(q) = 0$

2 $\varrho(p) = 0, \quad \varrho(q) = 1$

(c) $\varrho(p) = 1, \quad \varrho(q) = 0$

2 $\varrho(p) = 1, \quad \varrho(q) = 1$

4.) Which interpretations are models of the following set of formulae?

$$\{ (\neg q \equiv p), ((q \supset p) \wedge (q \equiv q)) \}$$

(a) $\varrho(p) = 0, \quad \varrho(q) = 0$

(b) $\varrho(p) = 0, \quad \varrho(q) = 1$

2 $\varrho(p) = 1, \quad \varrho(q) = 0$

(d) $\varrho(p) = 1, \quad \varrho(q) = 1$

5.) Define the semantic value of $(A \equiv B)$!

Score: 2 / 2.